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Topic: Rent a Bike Portal application

A Bike Rental Management System is a software application designed to facilitate the operation and management of a bike rental business through an online platform. This system provides tools for managing various aspects of the bike rental process, from bike inventory and customer bookings to payments and feedback. It aims to streamline operations, improve efficiency, and enhance the user experience for both administrators, bike owners and customers.

Modules:

1. Admin Module
2. Non- Registered User Module
3. Customer Module
4. Bike Owners Module

Project Name: Bike Rent Management System	
What? 1	How? 2
1. What are the modules required? Ans: a) Admin Module b) Non-Customer Module c) Customer Module d) Bike Owners Module	1) Admin: a) Login Method 1: Admin can login using name and password. Method 2: Admin can login using email id and password. Method 3: Admin can login using social media platforms. b) Admin can manage bikes: Method 1: Implement the manage brief information of every bike using bike name. Method 2: Implement the manage brief information of every bike using bike type. Method 3: Implement the manage brief information of every bike using bike id. c) Manage the Bike booking Application by admin: Method 1: Implement the view of the bike booking application using bike type. Method 2: Implement the view of the bike booking application using bike amount. Method 3: Implement the view of the bike booking application as a list. d) Admin can generate payment bill: Method 1: Generate bills based on rental duration and bike type. Method 2: Generate bills with itemized charges. Method 3: Generate bills using a template-based approach. e) View payment history by Admin: Method 1: View payment history by date range. Method 2: View payment history by customer. Method 3: View payment history in a summary report. f) Admin can review feedback from customers for bikes: Method 1: Review feedback as a list. Method 2: Review feedback by customer rating. Method 3: Review feedback by bike model. g) Admin can manage about us page: Method 1: Edit the About Us page with a react code.
2. What are the Functionalities required for Admin Module? Ans: Admin Module: a. Admin can login to the Dashboard. b. Admin can manage bikes. c. Admin can manage bike booking application. d. Admin can Generate payment bill. e. Admin can view payment history. f. Admin can review feedback from customers for bike. g. Admin can manage the About us page. h. Admin can manage customer support to users by addressing queries, resolving issues.	
3. What are the Functionalities required for Non-Registered User Module? Ans: Non-Registered User Module: a. Non- Registered User can browse for bikes. b. Non- Registered User can view bike details. c. Non- Registered User can view bike pricing details. d. Non- Registered User can register to dashboard.	
4. What are the Functionalities required for Customer Module? Ans: Customer Module: a. Customer can browse for bikes. b. Customer can view bike details. c. Customer can view bike pricing details. d. Customer can login to the dashboard. e. Customer can manage their personal information.	

- f. Customer can change their password.
- g. Customer can book bikes.
- h. Customer can make payment for bike bookings.
- i. Customer can view the payment history made for booking.
- j. Customer can give feedback or rating for bikes.

5. What are the Functionalities required for bike owner Module?

Ans: Bike Owner Module:

- a. Bike Owner can login to the dashboard.
- b. Bike Owner can add bike.
- c. Bike Owner can view, update, remove bikes.
- d. Bike Owner can view the payment received for their bikes.

Method 2: Edit the About Us page using a WYSIWYG editor.
Method 3: Edit the About Us page using HTML/CSS code.

h) Admin can manage customer support:

Method 1: Address queries via an email-based support system.

Method 2: Address queries through a live chat interface.

Method 3: Address queries through a ticketing system.

2) Non-Registered User:

a) Non- Registered User browse for bikes:

Method 1: Browse bikes using filters.

Method 2: Browse bikes using a search bar and filter.

Method 3: Browse bikes using a category view.

b) Non- Registered User can view the bike details:

Method 1: View details in a pop-up modal.

Method 2: View details in a detailed view page.

Method 3: View details in a side panel.

c) Non- Registered User can view bike pricing details:

Method 1: View pricing in a separate pricing section.

Method 2: View pricing on the bike detailed view page.

Method 3: View pricing in a comparison view.

d) Non- Registered User can Register to dashboard:

Method 1: Register using social media accounts.

Method 2: Register using a standard registration form.

Method 3: Register using a phone number.

3) Customer:

a) Customers browse for bikes:

Method 1: Browse bikes using filters.

Method 2: Browse bikes using a search bar and filter.

Method 3: Browse bikes using a category view.

b) Customers can view the bike details:

Method 1: View details in a pop-up modal.

Method 2: View details in a detailed view page.

Method 3: View details in a side panel.

c) Customers can view bike pricing details:

Method 1: View pricing in a separate pricing section.

Method 2: View pricing on the bike detailed view page.

Method 3: View pricing in a comparison view.

d) Customer can login to the dashboard:

Method 1: Customer can login using username and password.

Method 2: Customer can login using email id and password.

Method 3: Customer can login using social media platforms.

e) Customer can manage their personal information:

Method 1: Manage personal information with a form submission.

Method 2: Manage personal information using a settings panel.

Method 3: Manage personal information via a profile page.

f) Change the password by Customer:

Method 1: Customer can Reset the password using name.

	Method 2: Customer can Reset the password using email id. Method 3: Customer can Reset the password using social media platforms. g) Customer can book bike: Method 1: Book bikes through a booking form using bike name. Method 2: Book bikes through a booking form using bike variant. Method 3: Book bikes through a booking form using bike id. h) Customer can make payment for bike booking: Method 1: Make payment for bike booking using bike name. Method 2: Make payment for bike booking using bike variant. Method 3: Make payment for bike booking using bike id. i) Customer can view the payment history made for booking: Method 1: View payment history in a card. Method 2: View payment history in a section. Method 3: View payment history in a list and downloadable manner. j) Customer can give feedback or rating for bikes: Method 1: Submit feedback and rating through a feedback form using bike name. Method 2: Submit feedback and rating via a review section on bike details page. Method 3: Submit feedback and rating through a feedback form bike id. 4) Bike Owner: a) Bike Owner can login to the dashboard: Method 1: Bike Owner can login using username and password. Method 2: Bike Owner can login using email id and password. Method 3: Bike Owner can login using social media platforms. b) Bike Owner can Add bikes: Method 1: Add bike with bulk upload functionality. Method 2: Add bike using a detailed form. Method 3: Add bike using a guided wizard. c) Bike Owner can view, update and remove bike: Method 1: View, update, and remove bikes using bike name. Method 2: View, update, and remove bikes using bike id. Method 3: View, update, and remove bikes using bike category. d) Bike Owner can view the payment received for their bikes: Method 1: View payment history in a card. Method 2: View payment history in a section. Method 3: View payment history in a list and downloadable manner.
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1) Admin:

a) Login

Method 2: Admin can login using email id and password.

(If Admin login using email id and password, we can verify the originality of the user.)

b) Admin can manage bikes:

Method 3: Implement the manage brief information of every bike using bike id.

(Each bike has a unique ID, which avoids any ambiguity or confusion. You can directly access or manage a bike's information without worrying about duplicates. Bike IDs remain consistent even if other details about the bike change.)

c) Manage the Bike booking Application by admin:

Method 3: Implement the view of the bike booking application as a list.

(A list view provides a clear and organized way to present multiple bikes. Users can easily scroll through and see all available bikes, their details, and booking options. List views can be easily customized to include various columns or details such as bike ID, name, type, price, and availability.)

d) Admin can generate payment bill:

Method 1: Generate bills based on rental duration and bike type.

(This method is straightforward and easy to implement. Charges are based on the rental duration (e.g., hourly, daily) and the type of bike (e.g., mountain bike, road bike), which are fundamental elements of a rental system.)

e) View payment history by Admin:

Method 3: View payment history in a summary report.

(A summary report provides a consolidated view of payment history, offering key insights and trends at a glance. Instead of searching through individual records, a summary report presents aggregated data, saving time and reducing the need for detailed manual checks.)

f) Admin can review feedback from customers for bikes:

Method 1: Review feedback as a list.

(A list view provides a comprehensive overview of all feedback in one place. It allows you to see all comments, ratings, and other details briefly, making it easier.)

g) Admin can manage about us page:

Method 1: Edit the About Us page with a react code.

(React component-based architecture allows for modular and reusable code. This means you can manage different sections of the "About Us" page as separate components, making it easier to maintain.)

h) Admin can manage customer support:

Method 1: Address queries via an email-based support system.

(Email provides a written record of all communication. Email can handle a wide range of

1) Admin:

a) Login

1) Bike names might not be unique, leading to confusion if multiple bikes have the same or similar names. Bike type is useful for categorization but not for managing individual bike details or tracking.

b) Admin can manage bikes:

2) Presenting the view based solely on bike type may limit the user's ability to make a choice based on other important factors. Presenting a view based on bike amount (e.g., only showing bikes in a certain price range or quantity) might confuse users if the information isn't clear.

c) Manage the Bike booking Application by admin:

3) Viewing applications by bike type might be too broad. If you have multiple bike applications filtering by bike type alone might not provide enough granularity, especially if there are many applications within each type.

d) Admin can generate payment bill:

4) Itemizing every charge (e.g., base rate, additional fees, taxes) can complicate the billing process. Template-based billing might not accommodate all the variations and specifics of bike rental pricing.

e) View payment history by Admin:

5) While viewing by date range is useful for seeing transactions within a specific period, it doesn't provide a holistic view of overall payment trends or patterns. Viewing payment history by customer provides detailed information about individual customers but might not offer a broader perspective on overall payment.

f) Admin can review feedback from customers for bikes:

6) Focusing solely on customer ratings can miss valuable context provided in written feedback. While ratings provide a quick snapshot, they don't capture the nuances of customer experiences. Reviewing feedback by bike models can limit the view to specific models, which may not provide a complete picture of overall service quality.

g) Admin can manage about us page:

7) WYSIWYG (What You See Is What You Get) editors are great for basic content updates but may lack the flexibility required for more complex. HTML/CSS alone is static and does not easily accommodate dynamic content or interactions. If the page needs to be updated frequently or requires interactive features, HTML/CSS might be insufficient.

h) Admin can manage customer support:

8) Live chat requires real-time responses, which can be challenging to manage, especially during peak times or outside of business hours. For straightforward queries, a ticketing system might be more formal and cumbersome than necessary, potentially slowing down the support process.

2) Non- Registered User:

query types, including detailed descriptions, attachments (such as screenshots), and lengthy responses.)

2) Non- Registered User:

a) Non- Registered User browse for bikes:

Method 2: Browse bikes using a search bar and filter.

(Combining a search bar with filters allows users to quickly narrow down their options based on specific criteria. The search bar helps find bikes with specific keywords or attributes, while filters refine the results based on characteristics. This approach offers a dynamic browsing experience.)

b) Non- Registered User can view the bike details:

Method 2: View details in a detailed view page.

(A detailed view page allows for a full-screen, in-depth presentation of bike details. Users can see all relevant information, including high-resolution images, specifications, pricing, and availability.)

c) Non- Registered User can view bike pricing details:

Method 2: View pricing on the bike detailed view page.

(This ensures that users have all the necessary information in one place, making it easier to make a booking decision.)

d) Non- Registered User Can Register to dashboard:

Method 2: Register using a standard registration form.

(A standard registration form allows you to collect all necessary user information in a structured manner, including name, email, password, and any other details required for the rental system.)

3) Customer:

a) Customers browse for bikes:

Method 2: Browse bikes using a search bar and filter.

(Combining a search bar with filters allows users to quickly narrow down their options based on specific criteria. The search bar helps find bikes with specific keywords or attributes, while filters refine the results based on characteristics. This approach offers a dynamic browsing experience.)

b) Customers can view the bike details:

Method 2: View details in a detailed view page.

(A detailed view page allows for a full-screen, in-depth presentation of bike details. Users can see all relevant information, including high-resolution images, specifications, pricing, and availability.)

c) Customers can view bike pricing details:

Method 2: View pricing on the bike detailed view page.

(This ensures that users have all the necessary information in one place, making it easier to make a booking decision.)

d) Customers can login to dashboard:

Method 2: User can login using email id and password.

a) Non- Registered User browse for bikes:

1) While filters are excellent for refining options, they may not help users discover bikes if they don't know what specific attributes to filter by. Categorizing bikes might be limiting if the categories don't align with all the ways users might want to search or browse. This can lead to users feeling restricted to predefined categories.

b) Non- Registered User can view the bike details:

2) Pop-up modals have limited space, which can constrain the amount of information displayed. This can make it challenging to provide comprehensive details. While side panels can offer more space than modals, they still might not provide as much room as a full-page view.

c) Non- Registered User can view bike pricing details:

3) Having pricing information in a separate section means users have to navigate away from the detailed view of the bike to find out the cost. While comparison views can be useful for comparing multiple bikes, they might not provide detailed information about individual bike pricing.

d) Non- Registered User can Register to dashboard:

4) Users may have concerns about sharing their social media data with a third-party application, which can affect their willingness to register. Users might change their phone numbers more frequently, leading to potential issues with account recovery and communication.

2) Customer:

a) Customers browse for bikes:

1) While filters are excellent for refining options, they may not help users discover bikes if they don't know what specific attributes to filter by. Categorizing bikes might be limiting if the categories don't align with all the ways users might want to search or browse. This can lead to users feeling restricted to predefined categories.

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2) Pop-up modals have limited space, which can constrain the amount of information displayed. This can make it challenging to provide comprehensive details. While side panels can offer more space than modals, they still might not provide as much room as a full-page view.

c) Customers can view bike pricing details:

3) Having pricing information in a separate section means users have to navigate away from the detailed view of the bike to find out the cost. While comparison views can be useful for comparing multiple bikes, they might not provide detailed information about individual bike pricing.

d) Customers can login to dashboard:

5) To verify the originality of the user using a login method based solely on a username and password or social media for admin access might not be the most effective or secure approach for several reasons.

e) Customer can manage their personal information:

6) Using a form submission for managing personal information can be less user-friendly and may not

(If User login using email id and password, we can verify the originality of the user.)

e) Customer can manage their personal information:

Method 3: Manage personal information via a profile page.

(A profile page provides a dedicated and centralized location where users can view and update all their personal information. A profile page can be designed to offer a comprehensive and user-friendly interface. It allows users to easily access and modify their details without navigating through multiple forms.)

f) Change the password by Customer:

Method 2: User can Reset the password using email id.

(Allowing users to reset their password via email is a user-friendly approach)

g) Customer can book bike:

Method 3: Book bikes through a booking form using bike id.

(Bike IDs are unique identifiers for each bike. Using bike IDs ensures that the exact bike being booked is identified, which minimizes the risk of booking errors.)

h) Customer can make payment for bike booking:

Method 3: Make payment for bike booking using bike id.

(Bike IDs are unique identifiers for each bike. Using bike IDs for payment ensures that the exact bike being booked is identified, minimizing the risk of booking errors or confusion.)

i) Customer can view the payment history made for booking:

Method 3: View payment history in a list and downloadable manner.

(A list format allows users to view their payment history in a structured and organized way. Providing a downloadable format (such as PDF) allows users to keep a personal record of their payment history.)

j) Customer can give feedback or rating for bikes:

Method 3: Submit feedback and rating through a feedback form bike id.

(Bike IDs are unique identifiers assigned to each bike, ensuring that feedback and ratings are linked to the specific bike being reviewed. This precision helps in accurately associating feedback with the correct bike.)

4) Bike Owner:

a) Bike Owner can login to the dashboard:

Method 2: User can login using email id and password.

(If User login using email id and password, we can verify the originality of the user.)

b) Bike Owner can Add bikes:

Method 3: Add bike using a guided wizard.

(A guided wizard provides step-by-step instructions, making the process more accessible for users who might be unfamiliar with the system.)

c) Bike Owner can view, update and remove bike:

provide context or feedback about the user's current information. Settings panels are often used for managing preferences and system settings rather than personal information.

f) Change the password by Customer:

7) Tracking home bike applications is crucial for efficient management and customer service. Tracking by bike type alone does not provide detailed insights into other important aspects such as bike amount, applicant profile, or application status.

g) Customer can book bike:

8) Bike names may not be unique, especially if there are multiple bikes with similar or identical names. If a bike variant includes multiple bikes, there is a risk of errors in booking if users select a variant but intend to book a specific bike within that category. This method does not guarantee that the exact bike is reserved.

h) Customer can make payment for bike booking:

9) Bike names may not be unique, especially if there are multiple bikes of the same model or type. This can lead to confusion. Variants might include multiple bikes with similar characteristics. If payments are processed based on variants, there's a risk of confusion or errors.

i) Customer can view the payment history made for booking:

10) Card-based views can be restrictive in terms of space, making it challenging to display detailed transaction information in a concise manner. Sections might be static and less interactive compared to a list format.

j) Customer can give feedback or rating for bikes:

11) Bike names might not be unique, especially if there are multiple bikes of the same model or type. This can lead to confusion. A review section on the bike details page might be less effective if users need to navigate through multiple pages or sections to provide feedback.

4) Non-Customer:

a) Bike Owner can login to the dashboard:

1) To verify the originality of the user using a login method based solely on a username and password or social media for admin access might not be the most effective or secure approach for several reasons.

b) Bike Owner can Add bikes:

2) Bulk upload functionality is useful for adding large volumes of bikes but requires careful handling of data formats and files. Errors in the upload file can lead to data integrity issues or incorrect information being added. A detailed form can be overwhelming if it includes too many fields or options

c) Bike Owner can view, update and remove bike:

3) Bike names might not be unique, especially if there are multiple bikes with the same model or similar names. This can lead to confusion. If users need to perform operations on specific bikes within a category, managing by category alone might lead to inadvertent changes or

Method 2: View, update, and remove bikes using bike id.

(Bike IDs are unique identifiers assigned to each bike. This ensures that operations like viewing, updating, and removing bikes are accurately performed.)

d) Bike Owner can view the payment received for their bikes:

Method 3: View payment history in a list and downloadable manner.

(A list format provides a structured, detailed view of payment history. Users can easily see all transactions in chronological order. Offering a downloadable option (e.g., PDF) allows users to keep offline records of their payment history.)

Why? 3

removals affecting multiple bikes, increasing the risk of errors.

d) Bike Owner can view the payment received for their bikes:

4) Card-based views often have limited space, making it challenging to display detailed payment information effectively. A section view might not be as structured or organized as a list.

Why Not? 4